



Major Project

ECHO-D

Presented by:

Prince Yadav- R214222---

Rajvardhan Singh- R214222----

Mentored By- Dr Kamakshi Rautela



#RANKED 52
IN INDIA

#University Category



NO.1 PVT. UNIVERSITY IN
ACADEMIC REPUTATION IN INDIA



ACCREDITED **GRADE 'A'**
BY NAAC



PERFECT SCORE OF **150/150** AS A TESTAMENT
TO EXCEPTIONAL E-LEARNING METHODS

CONTENT

1. Introduction
2. Motivation
3. Problem Statement and Objective
4. Literature Review
5. Methodology
6. S.W.O.T Analysis
7. PERT Chart
8. Area of Application
9. Timeline & References



INTRODUCTION



- **What is Ultrasound?** A safe, non-invasive imaging technique using sound waves to visualize internal body structures.
- **The Challenge:** Manual interpretation of ultrasound images can be subjective, time-consuming, and highly dependent on operator experience.
- **The AI Solution:** Deep Learning (DL) models can automatically analyze ultrasound images to identify patterns, extract features, and predict clinical outcomes with high accuracy and speed.
- **Goal:** To enhance diagnostic accuracy, streamline workflows, and enable early disease detection.
- Medical ultrasound is a cornerstone of modern diagnostic imaging, valued across virtually every clinical specialty for its non-invasive nature, cost-effectiveness, and real-time capabilities. From monitoring fetal development in obstetrics to assessing cardiac function and guiding biopsies, its applications are vast and indispensable. However, the diagnostic power of ultrasound is fundamentally constrained by a critical challenge: the **subjectivity of human interpretation**.
- The accuracy of a diagnosis often hinges on the skill and experience of the operator, leading to significant variability between practitioners and even for the same practitioner over time. This reliance on qualitative assessment means that subtle, yet diagnostically crucial, patterns within the sonographic data may be overlooked.



MOTIVATION

The core motivation is to **transform patient outcomes** by empowering clinicians with intelligent tools. We aim to shift the paradigm from reactive treatment to proactive, predictive healthcare.

- Ultrasound is widely used in clinical practice due to its **non-invasive, affordable, and real-time imaging** capabilities.
- Its effectiveness is limited by **subjective human interpretation**, which depends on the operator's expertise.
- This subjectivity causes **variability in diagnostic accuracy** across practitioners and over time.
- **Subtle yet critical patterns** in ultrasound images are often missed during manual evaluation.
- There is a strong need for **automated, objective, and reliable methods**—such as machine learning—to enhance diagnostic accuracy and consistency.



PROBLEM STATEMENT

- **Problem Statement:** The diagnostic accuracy of ultrasound is fundamentally limited by technical artifacts, such as **acoustic shadowing**, and by subjective operator interpretation. This unreliability poses a significant risk of misdiagnosis and delayed treatment, which negatively impacts patient outcomes
- **Primary Objective:** To develop and validate a deep learning framework that automates the analysis of ultrasound images for:
 - **Knowledge Extraction:** Accurately segmenting organs and identifying anomalies
 - **Outcome Prediction:** Forecasting disease progression, treatment response, or patient risk.

OBJECTIVE

- To design, develop, and validate an end-to-end deep learning system that automatically extracts critical knowledge from ultrasound images and accurately predicts clinical outcomes, thereby enhancing diagnostic precision and supporting clinical decision-making.
- the objective is to build and validate an AI that analyzes ultrasound images to automatically identify diseases and predict patient outcomes, creating a powerful decision-support tool for clinicians.

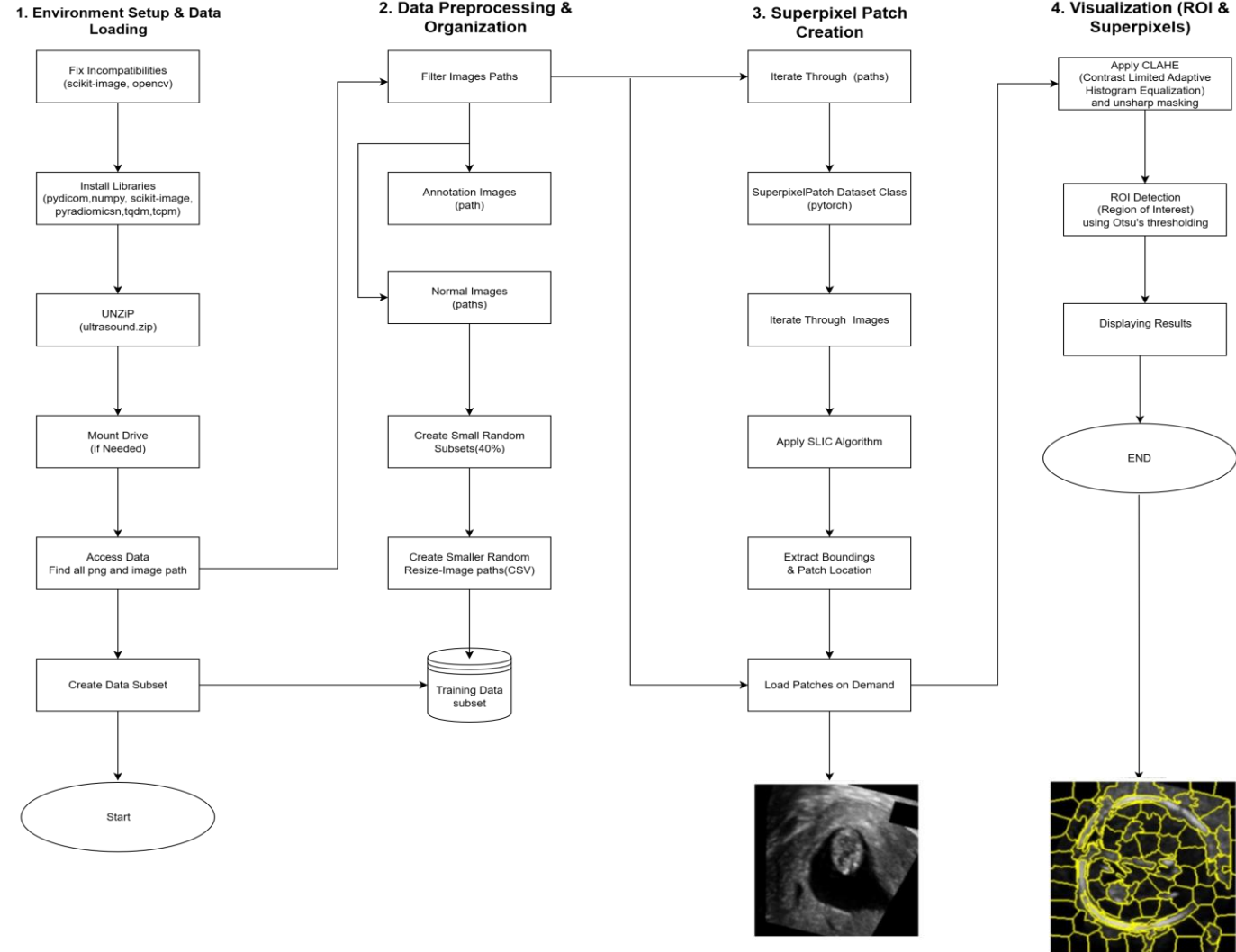
LITERATURE REVIEW



No.	Author(s)	Title / Source	Key Contribution	Relevance
1	G. J. Burton, C. P. Sibley, E. R. Jauniaux	Placental Anatomy and Physiology, 2016	Explains placenta structure and physiology	Guides image interpretation and segmentation
2	A. Arakaza, L. Zou, J. Zhu	Placenta Accreta Spectrum Diagnosis Challenges, 2023	Reviews PAS diagnostic challenges	Motivates AI-based preprocessing
3	H. Wang et al.	DL Pipeline for PAS Evaluation with MRI, 2024	Uses medical priors in DL pipeline	Aligns with preprocessing and segmentation
4	D. Young et al.	Ultrasound Texture Feature Fusion, 2024	Applies ML on ultrasound features	Demonstrates relevant feature extraction
5	E. Jauniaux, S. Collins, G. J. Burton	PAS Anatomy for Prenatal Ultrasound, 2018	Links anatomy with imaging markers	Defines ROI for segmentation
6	J. Morales-Roselló et al.	Ultrasound vs MRI in PAS, 2021	Compares imaging modalities	Supports ultrasound preprocessing

METHODOLOGY

- **Step 1 – Image Enhancement:**
 - Applied **CLAHE** for local contrast improvement.
 - Used **Unsharp Masking** to sharpen edges and highlight structures.
- **Step 2 – ROI Detection:**
 - Converted to grayscale and used **Otsu's Thresholding** to isolate the main object.
 - Cropped the **Region of Interest (ROI)** using a bounding box.
- **Step 3 – Superpixel Segmentation:**
 - Applied **SLIC algorithm** to divide ROI into ~300 meaningful superpixels.
 - Simplified image structure for efficient analysis.

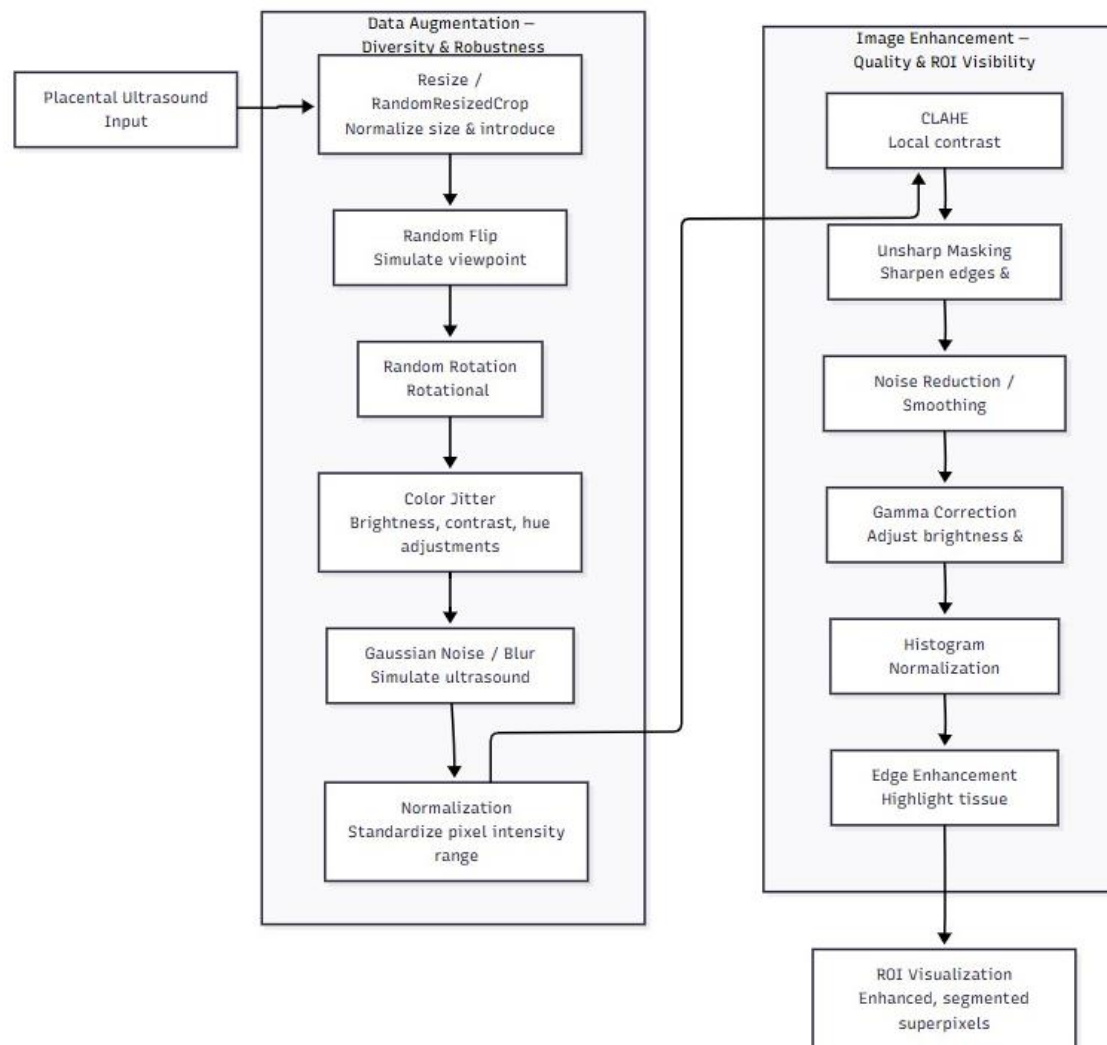


METHODOLOGY

- **Superpixel Patch Dataset:**
 - Created small image patches from superpixels using a custom PyTorch dataset.
 - Applied data augmentation (resize, flip, color jitter) for diversity.
- **Feature Extraction & Clustering:**
 - Extracted deep features using a pre-trained CNN.
 - Performed unsupervised clustering to group visually similar regions.

Outcome:

- Automatic pattern discovery in ultrasound images.
- Enables faster, more accurate analysis with reduced data complexity.

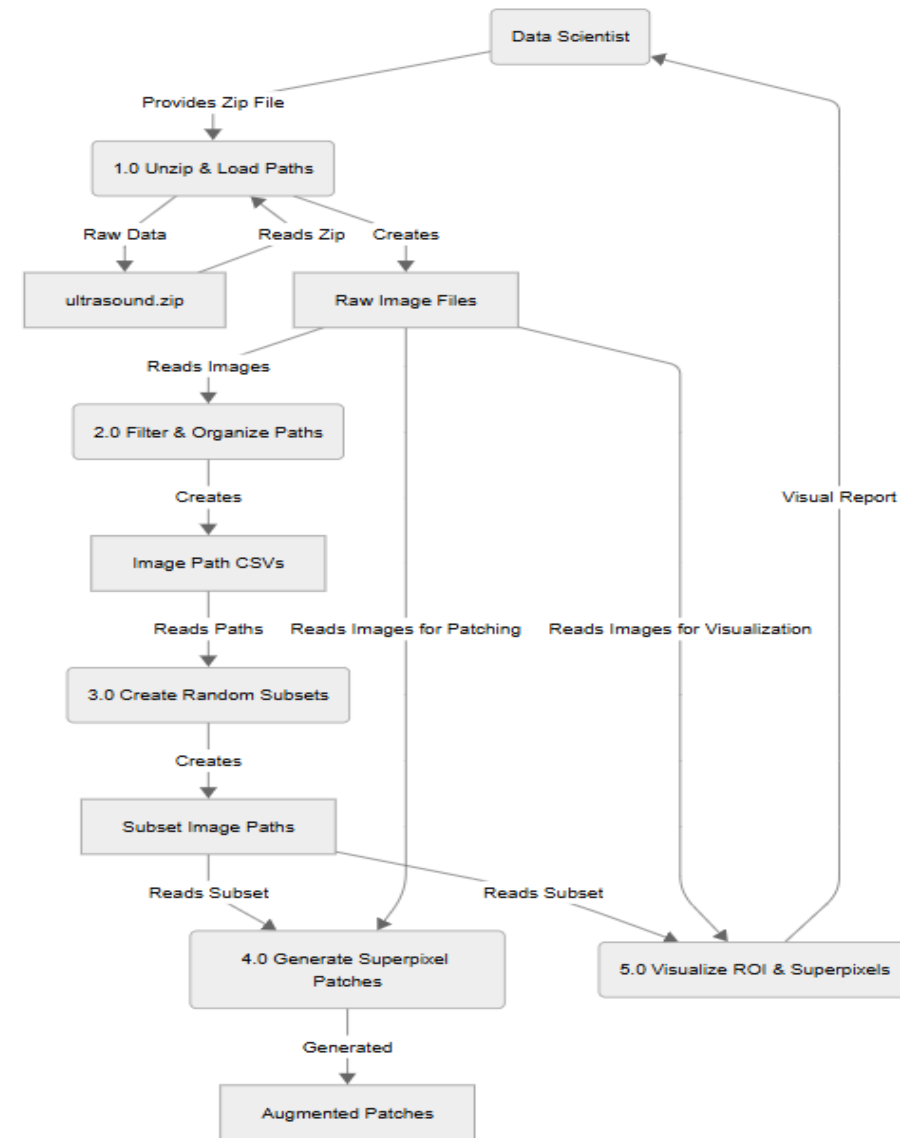


WORKING MODLE

Technical Diagram and Working Module

Superpixel Patch Extraction & Visualization Pipeline

- Pipeline Overview:
 - Input ultrasound image dataset (ZIP) is received and unpacked.
 - Files are filtered and organized into annotation and image sets.
 - Randomized subsets are created for robust training/analysis.
 - Superpixel segmentation (SLIC algorithm) is applied to generate localized patches for deep learning tasks.
- The pipeline supports integration with Google Colab for data loading, augmentation, and patch extraction.
 - Visualization utilizes CLIHE & Otsu's Thresholding for ROI and superpixel boundary display.
- Working Modules:
 - Data Loading & Preprocessing: Unzipping, CSV path generation, subset selection.
 - Patch Generation Module: Applies SLIC, extracts and organizes patch data.
 - Visualization: Processes ROI, overlays superpixel regions on extracted images.
 - Environment runs via Colab with PyTorch, OpenCV, pandas, scikit-image, and visualization libraries.



SWOT ANALYSIS



S

STRENGTHS

Provides faster, more accurate, and objective diagnoses than human interpretation alone.



W

WEAKNESSES

internal characteristics that put the company at a disadvantage relative to others



O

OPPORTUNITIES

Can be integrated with portable devices for remote and point-of-care diagnostics (telemedicine).

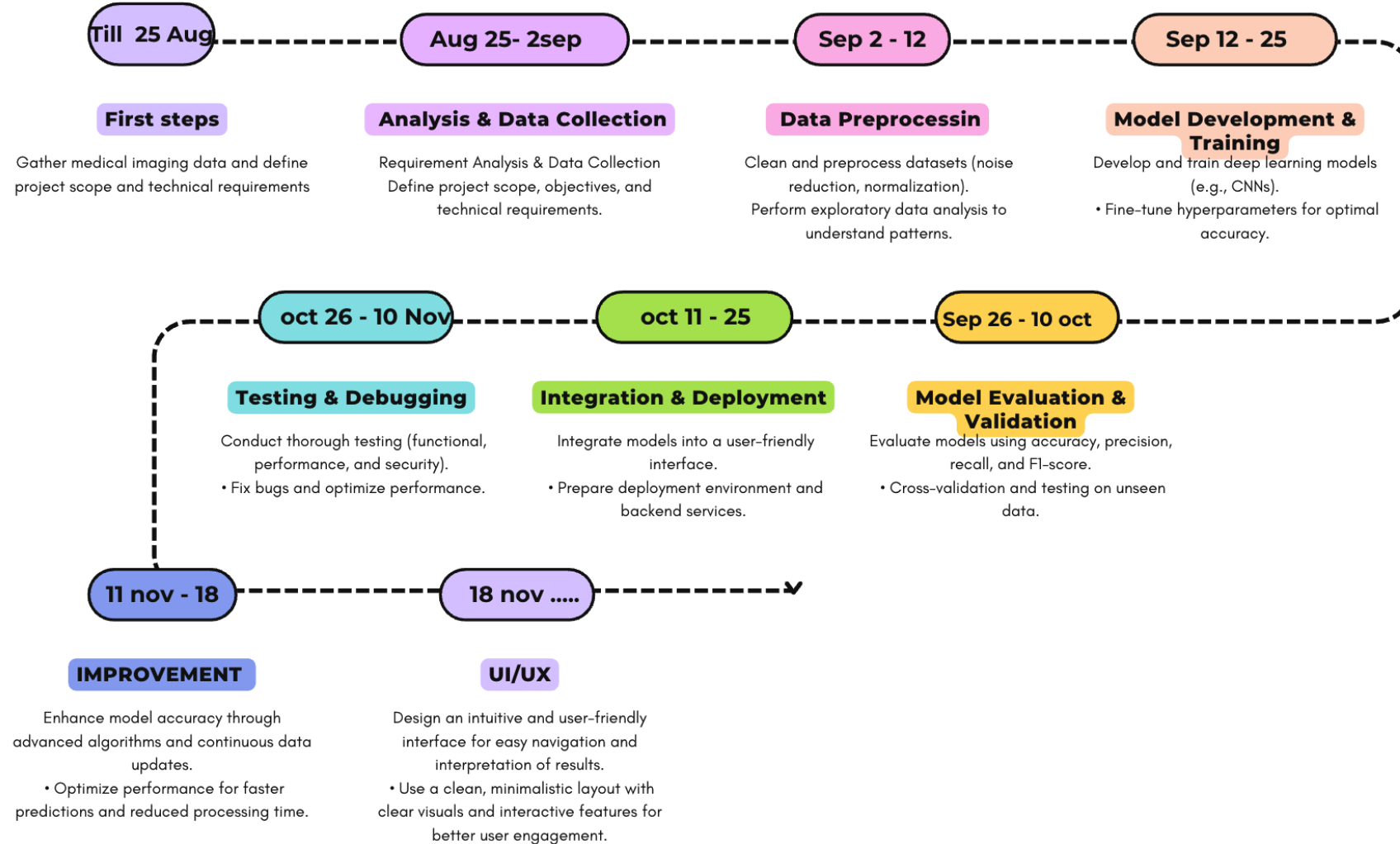


T

THREATS

Faces significant regulatory hurdles (e.g., FDA approval) and potential resistance from clinicians.

Timeline



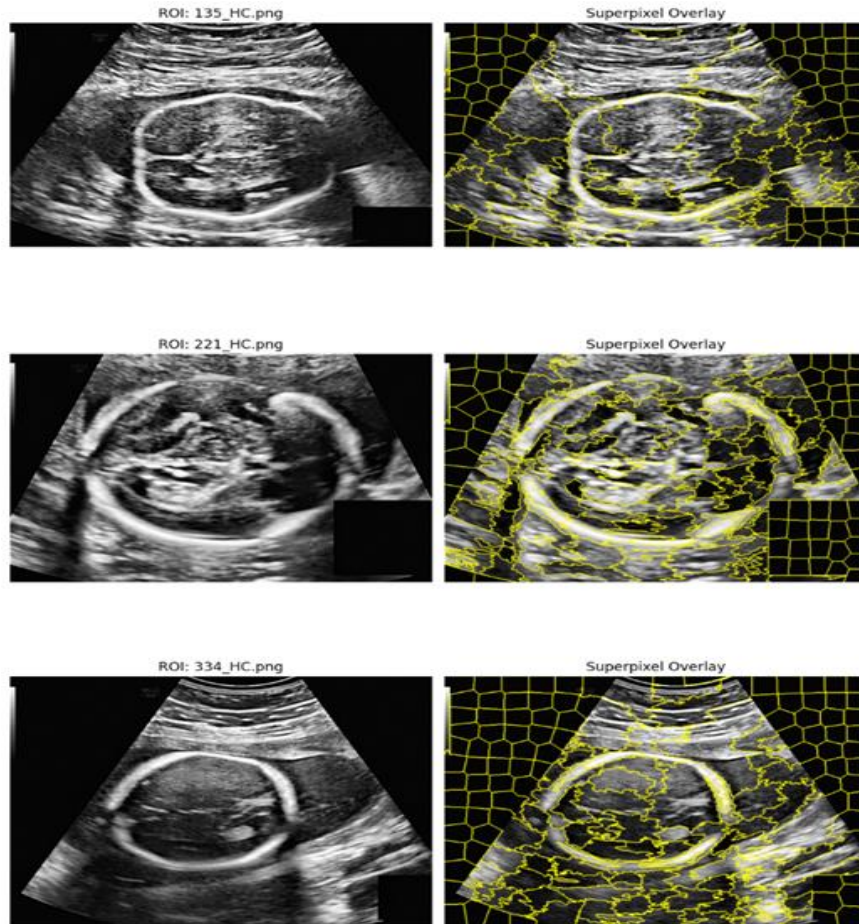
AREA OF APPLICATION

- **Obstetrics & Gynecology:** Fetal growth monitoring, anomaly detection, and assessing ovarian/uterine health.
- **Cardiology (Echocardiography):** Automatic measurement of heart function (ejection fraction), and valve disease detection.
- **Oncology:** Differentiating benign vs. malignant tumors in breast, thyroid, liver, and monitoring treatment response.
- **Radiology:** Assisting in the characterization of lesions in the liver, kidneys, and other organs.
- **Point-of-Care Ultrasound (POCUS):** Guiding needle placements and enabling rapid diagnosis in emergency settings.

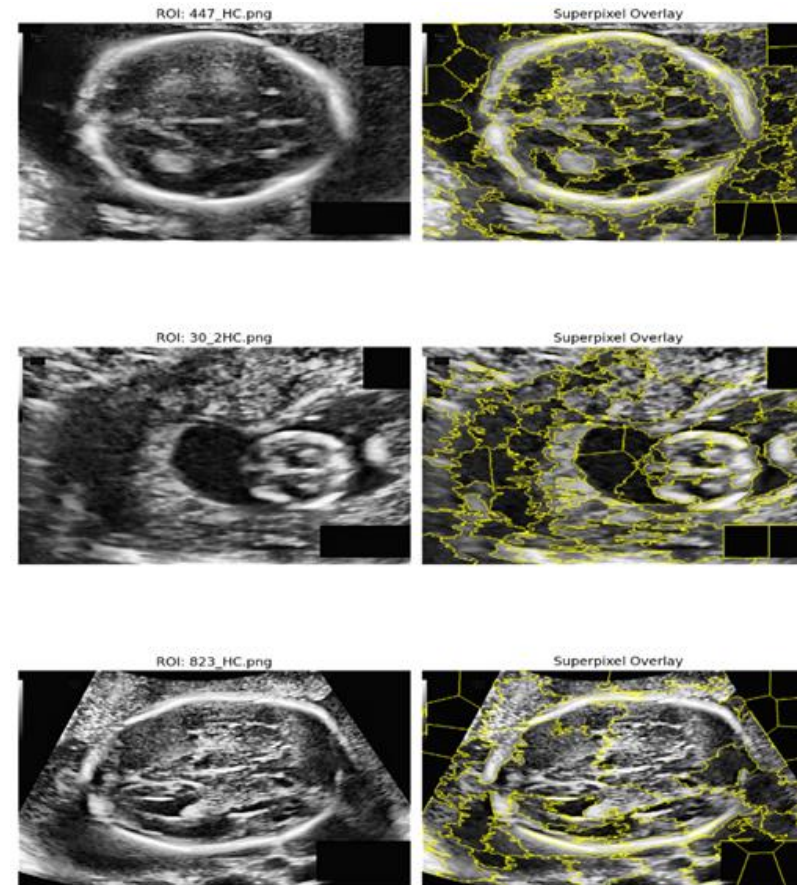


RESULTS

THEN

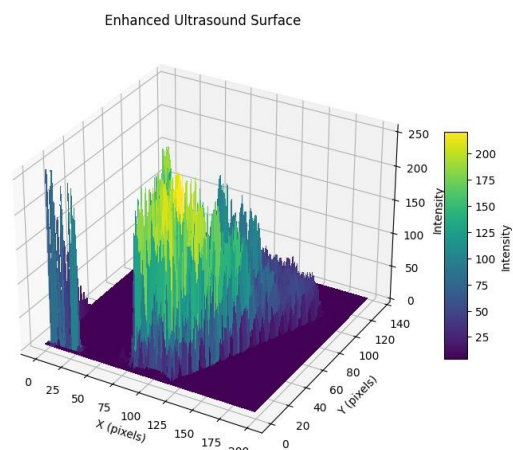
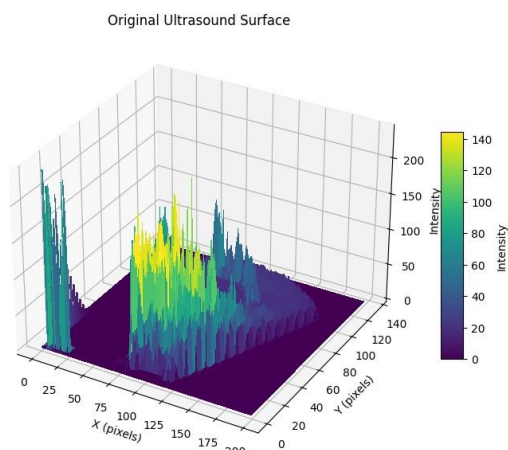


NOW



RESULTS

```
--2025-12-07 12:21:38-- https://dl.fbaipublicfiles.com/segment\_anything/sam\_vit\_b\_01ec64.pth  
Resolving dl.fbaipublicfiles.com (dl.fbaipublicfiles.com)... 13.35.37.111, 13.35.37.123, 13.35.37.90, ...  
Connecting to dl.fbaipublicfiles.com (dl.fbaipublicfiles.com)[13.35.37.111]:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 375042383 (358M) [binary/octet-stream]  
Saving to: '/content/checkpoints/sam_vit_b_01ec64.pth'  
  
sam_vit_b_01ec64.pt 100%[=====>] 357.67M  191MB/s  in 1.9s  
  
2025-12-07 12:21:39 (191 MB/s) - '/content/checkpoints/sam_vit_b_01ec64.pth' saved [375042383/375042383]
```



RESULTS

- **SAM** generates high-precision anatomical segmentation masks.
- **BLIP-2** creates radiologist-style medical captions from ROI.
- **Grad-CAM++** produces smooth, high-resolution interpretability heatmaps.
- **Synthetic generator** creates ultrasound-like images to expand the dataset.

```
prompt = """
You are a medical imaging expert.

Describe this ultrasound image in a detailed and complete manner.
Include:
- Overall image appearance
- Presence or absence of lesions
- Size, shape, and margins of any observed lesion
- Echogenicity and texture
- Whether the findings suggest normal, benign, or malignant characteristics

Write the description as a formal medical caption.
"""

inputs = processor(
    images=image,
    text=prompt,
    return_tensors="pt"
).to(model.device, torch.float16)

with torch.no_grad():
    output = model.generate(
        **inputs,
        max_new_tokens=80
    )

caption = processor.decode(output[0], skip_special_tokens=True)
print(f"Medical Caption: {caption}")

Medical Caption: a uterine fibroid is a benign tumor that is found in the uterus and is a common cause of uterine cancer
```

CAPTION:

Segmented ultrasound region of interest.

FINDINGS:

- The highlighted ROI demonstrates grayscale echotexture confined to the segmentation boundaries.
- The internal echogenicity appears relatively homogeneous on this static image.
- No distinct cystic component, calcification, or sharply marginated focal lesion is appreciable.
- Margins correspond to the applied segmentation mask.

IMPRESSION:

- The appearance of the segmented region is nonspecific and appears within expected limits on this limited evaluation.
- Subtle pathology cannot be excluded without additional imaging planes or dynamic assessment.

LIMITATIONS:

- Analysis based on a single static ultrasound image.
- Limited field of view confined to the segmented ROI.
- No Doppler or quantitative measurements available.

References

- Acharya, U. R., Faust, O., & Ng, E. Y. K. (2010). *Ultrasound imaging: Medical image processing and analysis techniques*. *Journal of Medical Systems*, 34(5), 803–810. [1]
- Shorten, C., & Khoshgoftaar, T. M. (2019). *A survey on image data augmentation for deep learning*. *Journal of Big Data*, 6(1), 60. [2]
- Gonzalez, R. C., & Woods, R. E. (2018). *Digital image processing* (4th ed.). Pearson.
- Zhang, Y., Liu, S., & Wang, X. (2021). *Medical image enhancement techniques: A review*. *Computers in Biology and Medicine*, 133, 104373. [3]
- Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., ... van Ginneken, B. (2017). *A survey on deep learning in medical image analysis*. *Medical Image Analysis*, 42, 60–88. [4]
- Otsu, N. (1979). *A threshold selection method from gray-level histograms*. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62–66. [5]
- Pluim, J. P., Maintz, J. B., & Viergever, M. A. (2003). *Mutual-information-based registration of medical images: A survey*. *IEEE Transactions on Medical Imaging*, 22(8), 986–1004. [6]
- Rundo, L., Militello, C., Vitabile, S., & Battiato, S. (2018). *Medical image segmentation and enhancement techniques: A survey*. *Journal of Imaging*, 4(9), 110. [7]
- Shen, D., Wu, G., & Suk, H.-I. (2017). *Deep learning in medical image analysis*. *Annual Review of Biomedical Engineering*, 19, 221–248. [8]
- Perez, L., & Wang, J. (2017). *The effectiveness of data augmentation in image classification using deep learning*. *arXiv preprint arXiv:1712.04621*.
- Ravi, D., Wong, C., Deligianni, F., Berthelot, M., Andreu-Perez, J., Lo, B., & Yang, G.-Z. (2017). *Deep learning for health informatics*. *IEEE Journal of Biomedical and Health Informatics*, 21(1), 4–21. [9]
- Jiang, S., Xue, H., & Shen, D. (2018). *Placenta segmentation in ultrasound images: A review*. *Computers in Biology and Medicine*, 96, 123–133. [10]



THANK YOU

